

Python for AI Workshop

Installing uv

uv is a fast Python package and project manager written in Rust. It can be installed on Windows using a standalone installer via PowerShell.

1. Open **PowerShell** as an administrator. You can do this by searching for "PowerShell" in the Start menu, right-clicking the application, and selecting **Run as administrator**.
 2. In the PowerShell window, run the following command to download and execute the installer script:
`powershell -ExecutionPolicy Bypass -c "irm https://astral.sh/uv/install.ps1 | iex"`
 3. Once the installation is complete, close the PowerShell window and open a new one to ensure that the uv command is available in your system's PATH.
 4. Verify the installation by running:
`uv --version`
This command should display the installed version of uv.
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Installing Git

You can install Git for Windows using either the winget command-line tool or the standalone installer.

Method 1: Install using winget (Recommended)

winget, or the Windows Package Manager, is a command-line utility for installing applications on Windows. It's pre-installed on modern versions of Windows 10 and 11.

1. Open **Command Prompt** or **PowerShell**.
2. Run the following command to install Git:
`winget install --id Git.Git -e --source winget`
3. The installer will run and complete the installation.

Method 2: Alternative Installation via Git Installer

If winget doesn't work or you prefer a graphical installer, you can download the official Git for Windows installer.

1. Navigate to the official Git website's download page: <https://git-scm.com/download/win>

2. The download should begin automatically. If not, click on the "Click here to download" link for the latest version.
 3. Once the download is complete, run the installer and follow the on-screen prompts.
 - Accept the default options during installation unless you have a specific reason to change them.
 - It's generally recommended to keep the default settings for line ending conversions and the Git editor.
 4. Once the installation is finished, open **Git Bash** (which is installed alongside Git) to verify the installation. You can also use Command Prompt or PowerShell.
 5. Run the following command:
`git --version`
This will show the installed Git version.
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Configuring Git

After installing Git, you should configure your user name, email, and preferred text editor. These settings are stored globally and will apply to all your Git repositories on the computer.

Set User Name and Email

Your user name and email are embedded in every commit you make, identifying you as the author.

1. Open **Git Bash**, **Command Prompt**, or **PowerShell**.
2. Set your user name with the following command, replacing "Your Name" with your actual name:
`git config --global user.name "Your Name"`
3. Set your email address with this command, replacing "your.email@example.com" with your actual email:
`git config --global user.email "your.email@example.com"`

Set Core Editor

Git needs a text editor to handle tasks like writing commit messages or rebasing. By default, it often uses Vim, which can be difficult for beginners. You can change this to a more user-friendly editor such as Visual Studio Code (VS Code).

To set VS Code as the default editor, run:

```
git config --global core.editor "code --wait"
```

- The --wait flag is crucial. It tells Git to wait for the editor to close before continuing the Git process. This is necessary because Git needs to know when you are finished editing a commit message, for example.

Verify Your Settings

To confirm that your settings have been configured correctly, you can run:

```
git config --list --global
```

This command will display all your global Git configurations, including the user name, email, and core editor.